

CHAPTER TWO

LITERATURE REVIEW

2.1 Introduction

- ◆ **Theoretical Framework:** This section will review the relevant theories on attitudes and beliefs, including the Theory of Planned Behavior and Self-Efficacy Theory.
- ◆ **Previous Studies:** A review of previous research on the impact of teachers' attitudes and beliefs on instructional practices, specifically in mathematics education.
- ◆ **Problem-Solving Strategies in Mathematics:** An overview of effective problem-solving strategies and their importance in secondary education.
- ◆ **Contextual Factors in Sri Lanka:** A discussion on the educational context in Sri Lanka, including cultural, societal, and institutional influences on teaching practices.

This chapter reviews relevant literature related to the use of problem-solving strategies in mathematics education and the competency of teachers in implementing these strategies. It examines theoretical perspectives, empirical studies, and challenges associated with teaching problem-solving in mathematics, with special attention to the Sri Lankan context.

2.2 Concept of Problem Solving in Mathematics

Problem solving is a central component of mathematics learning. It involves the ability to apply mathematical knowledge and skills to unfamiliar situations in order to find solutions. According to George Pólya, problem solving is not merely about obtaining answers but understanding the process of arriving at solutions.

Pólya proposed four fundamental steps in problem solving:

1. Understanding the problem
2. Devising a plan
3. Carrying out the plan
4. Looking back (reflection)

These steps have been widely adopted in mathematics education and serve as a foundation for teaching problem-solving strategies.

2.3 Problem-Solving Strategies in Mathematics Teaching

Problem-solving strategies are structured approaches that help learners tackle mathematical problems effectively. Common strategies include:

- Drawing diagrams or visual representations
- Identifying patterns
- Working backward
- Making systematic lists
- Using logical reasoning

Research shows that students who are explicitly taught these strategies perform better in solving non-routine problems. Effective teaching of these strategies requires teachers to model thinking processes, encourage discussion, and create opportunities for exploration.

2.4 Teacher Competency in Mathematics Instruction

Teacher competency refers to the knowledge, skills, and attitudes required to facilitate effective learning. In mathematics education, this includes:

- Content knowledge
- Pedagogical knowledge
- Ability to use appropriate teaching strategies
- Classroom management skills

Competent teachers are able to adapt their teaching methods to suit students' needs and promote higher-order thinking skills. The integration of problem-solving strategies into teaching requires a deep understanding of both mathematics content and pedagogy.

2.5 Theoretical Framework

This study is grounded in constructivist learning theory, which emphasizes that learners actively construct knowledge through experience and interaction.

Key contributors to this theory include Jean Piaget and Lev Vygotsky.

- **Piaget's Theory:** Suggests that learners build knowledge through stages of cognitive development.
- **Vygotsky's Theory:** Highlights the importance of social interaction and the Zone of Proximal Development (ZPD), where learners can achieve higher understanding with guidance.

Problem-solving strategies align with these theories as they encourage active engagement, collaboration, and critical thinking.

2.6 Empirical Studies on Problem-Solving in Mathematics

Several studies have examined the impact of problem-solving strategies on mathematics learning:

- Studies indicate that students exposed to problem-solving-based instruction show improved conceptual understanding and retention.
- Research also shows that teacher training programs significantly enhance the use of problem-solving strategies in classrooms.
- Some studies highlight that teachers often lack sufficient training in implementing these strategies effectively.

In developing countries, including Sri Lanka, challenges such as limited resources, large class sizes, and exam-oriented teaching further hinder the use of problem-solving approaches.

2.7 Challenges in Implementing Problem-Solving Strategies

Teachers face several challenges when integrating problem-solving strategies into their teaching:

- **Lack of Professional Training:** Many teachers have not received adequate training in modern teaching methods.
- **Time Constraints:** Completing the syllabus often takes priority over in-depth problem-solving activities.
- **Examination Pressure:** Focus on high-stakes exams encourages rote learning.
- **Limited Resources:** Insufficient teaching materials and support.
- **Student Factors:** Students may lack foundational knowledge or motivation.

These challenges highlight the need for targeted interventions to enhance teacher competency.

2.8 Mathematics Education in Sri Lanka

In Sri Lanka, mathematics is a compulsory subject up to the General Certificate of Education (GCE) Ordinary Level. Despite curriculum reforms aimed at promoting student-centered learning, traditional teaching methods still dominate many classrooms.

The National Institute of Education Sri Lanka has introduced guidelines encouraging the use of activity-based and problem-solving approaches. However, implementation remains inconsistent due to gaps in teacher training and support.

2.9 Strategies for Enhancing Teacher Competency

Improving teacher competency requires a multifaceted approach:

- **Professional Development Programs:** Workshops and training sessions on problem-solving strategies.
- **Collaborative Learning:** Peer discussions and sharing best practices.
- **Use of Technology:** Incorporating digital tools to support teaching.
- **Reflective Practice:** Encouraging teachers to evaluate and improve their teaching methods.

Continuous professional development is essential for sustaining improvements in teaching practices.

2.10 Summary of the Literature Review

The literature highlights the importance of problem-solving strategies in mathematics education and the critical role of teacher competency in implementing these strategies effectively. While theoretical and empirical evidence supports the use of problem-solving approaches, challenges remain, particularly in contexts like Sri Lanka.

This study seeks to address these gaps by developing and evaluating an intervention aimed at enhancing teachers' competency in using problem-solving strategies.